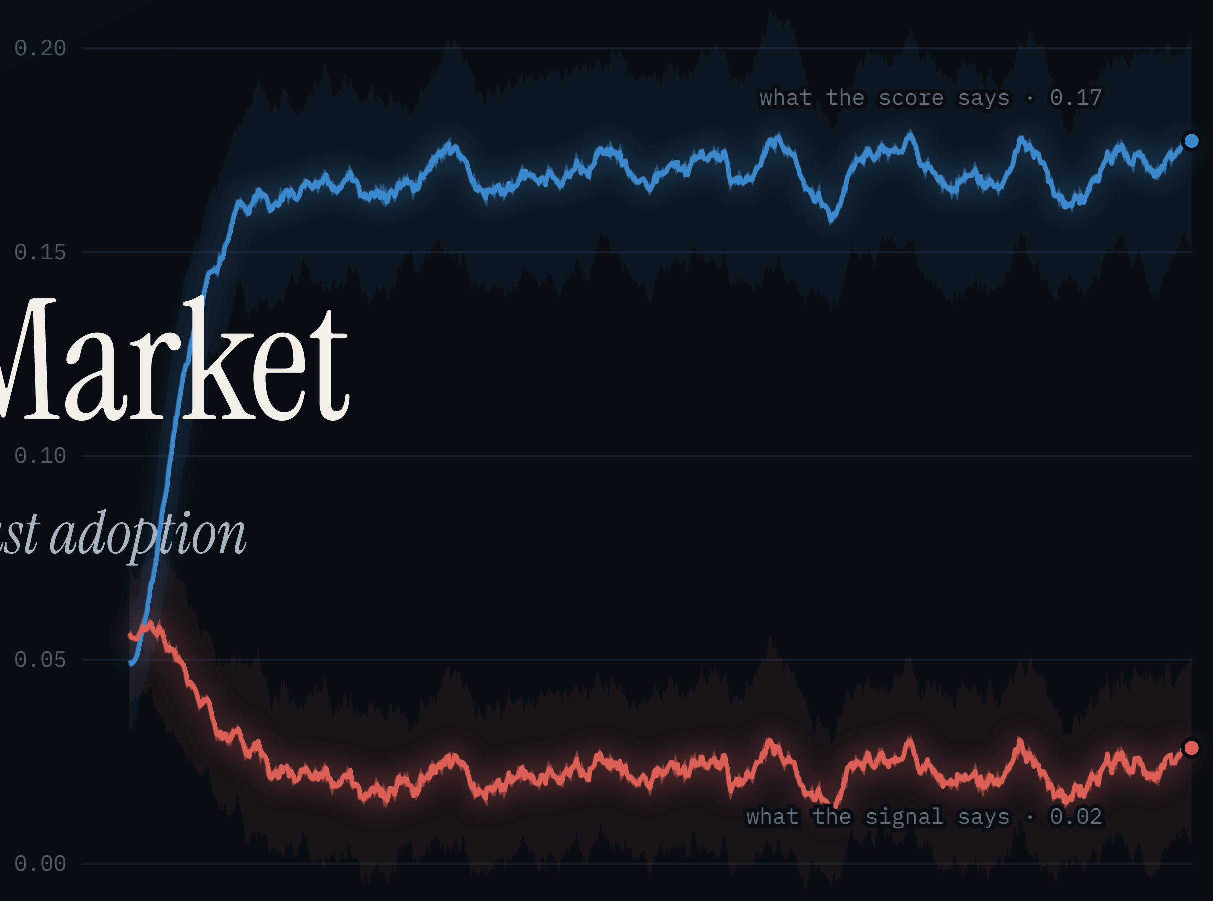
MSC CAPSTONE DEFENSE · IE SCHOOL OF SCIENCE & TECHNOLOGY

Performative Prediction in a Single-Asset Market

Self-fulfilment and signal erosion under forecast adoption



one forecast · two readings · 50 seeds, 30,000 periods

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MSc in Business Analytics & Data Science · June 2026

AGENT-BASED SIMULATION · 3,600-RUN PARAMETER SWEEP

→ TO BEGIN

ACT I · THE PROBLEM

When a forecast gains users, its success starts to move its own target

T_0 · LAUNCH

Nobody trades on it

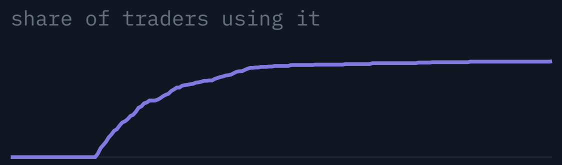
A service publishes a one-step return forecast. Its accuracy measures whatever **genuine structure** the rule has found.

→

ADOPTION

Subscribers act on it

Users trade in the direction of the signal. Their **coordinated orders move the price** the forecast is scored against.



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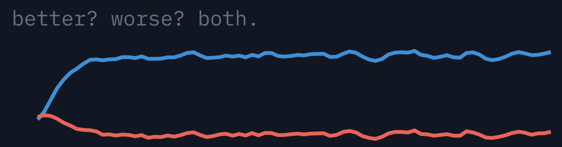
THE TRACKED SCORE SHIFTS

What exactly changed?

The returns now **carry the forecast's own fingerprint**. The score moved, but the reading is no longer innocent.



The forecast's skill, the market, *or the meaning of the measurement?*

Three literatures meet here, and none scores a forecast *inside the market it moves*

TRADITION 01

Reflexivity & performative prediction

Views alter the outcomes they are about. Deployment shifts the distribution a model is judged on; accuracy becomes a property of model plus behavioural response.

Soros 1987 · Perdomo et al. 2020

TRADITION 02

Heterogeneous-agent finance

Mean-variance traders switch between competing rules; a market maker moves the price linearly in net order flow. Supplies a market where trading feeds back into returns.

Brock & Hommes 1998 · Beja & Goldman 1980 · Farmer & Joshi 2002

TRADITION 03

Out-of-sample forecast evaluation

Fit through t, forecast $t+1$, summarise as out-of-sample R^2. The machinery presumes a target the forecast does not move.

Diebold & Mariano 1995

Adoption breaks exactly the premise the evaluation tradition rests on. *That intersection is unoccupied.*

The smallest market where the question can be posed, *and answered, exactly*

CONTRIBUTION 01

A performative forecast inside a real market model

A shared autoregressive rule is traded by its own users in a Brock-Hommes market with linear price impact: the feedback channel is a single explicit term.

CONTRIBUTION 02

Two scoreboards on the same path

The same forecast is scored against the realised return and against a counterfactual with the users' price impact removed, so self-fulfilment and erosion can be separated cleanly.

CONTRIBUTION 03

A quantified map, not an anecdote

A 3,600-run sweep over price impact, persistence, window and AR order locates the adoption share at which the independent signal has lost half its content.

CONTRIBUTION 04

Fully reproducible laboratory

Pure-numpy simulator, every figure regenerable from a seed, 43-test suite, CI on every push. Every number in this talk traces to a committed notebook.

How does adoption of a shared return forecast change its measured accuracy, *and the independent signal that accuracy is supposed to certify?*

SELF-FULFILLING CHANNEL ↗

Accuracy against the **realised return**: the only reading a deployed forecaster can observe. Expected to **rise** with adoption.

EROSION CHANNEL ↘

Accuracy against the **demand-adjusted counterfactual**: the path stripped of the users' own price impact. Expected to **fall**.

One equation carries the whole mechanism

$$r_{t+1} = \varphi\, r_t + \mu\, D_t + \sigma\, \varepsilon_{t+1}$$

φ r: the signal

A small persistent component, **φ = 0.25** at baseline. This is the structure the forecasting rule is built to recover.

μ D: the feedback door

A market maker absorbs aggregate demand and moves the quote. **The only channel through which trading re-enters returns.**

σ ε: the news

Exogenous Gaussian shocks. Unpredictable by construction; no forecast earns anything here.

N = 200 traders

submit signed orders q_i each period

Market maker

absorbs mean demand D, shifts log-price by μD

Return realises

signal + impact + news; next period repeats

N = 200 | T = 8,000 | μ = 0.05 | φ = 0.25 | σ = 0.01 | metrics on trailing 1,000

Two rules trade; adoption is the dial we turn

NON-ADOPTERS · THE NULL RULE

Active but skill-free

Zero-mean random orders: they keep the market trading and inject **directionally uninformative flow**. The benchmark any real signal must beat.

orders: zero-mean noise

ADOPTERS · THE SHARED RULE

One public AR(1) forecast

Rolling least-squares on the last **w = 250** returns, strictly out of sample. Mean-variance position, capped: adopters become **fixed-size traders on the forecast's sign**. Every adopter sees the same number.

orders: fixed size, forecast sign (capped)

HOW ADOPTION SPREADS

MECHANISM A · EXOGENOUS

Stochastic diffusion

Each non-adopter discovers the rule with probability π per period. Adoption is **imposed**, isolating the pure effect of wider use.

MECHANISM B · ENDOGENOUS

Performance-based switching

Entry follows recent **risk-adjusted profit** of the rule versus the null (logistic in the certainty-equivalent gap). Early success recruits, recruits reshape returns.

Score the same forecast on two targets: *with, and without, its users' own impact*

r_{t+1} =	$\varphi\, r_t$ · signal	$\mu\, D_t$ · users' impact	$\sigma\, \varepsilon$ · news	realised target
x_{t+1} =	$\varphi\, r_t$ · signal	$\mu\, D_t$ removed	$\sigma\, \varepsilon$ · news	demand-adjusted target

SCOREBOARD 1 · REALISED

What a deployed forecaster sees

R^2_{realised} : *forecast vs* r_{t+1}

The return the market prints, **price impact included**. The only score computable from outside.

expected to **rise** with adoption

SCOREBOARD 2 · DEMAND-ADJUSTED

Did the independent signal survive?

R^2_{da} : *forecast vs* $x_{t+1} = r_{t+1} - \mu D_t$

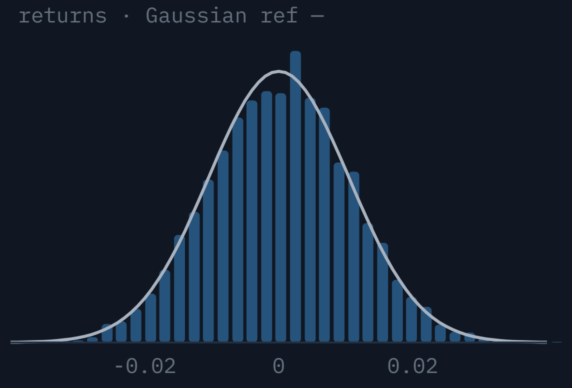
A counterfactual: the path had adopters **not moved today's quote**. Keeps the signal, strips the self-fulfilment.

expected to **fall** with adoption

The laboratory is clean before the feedback valve opens

Baseline market behaves

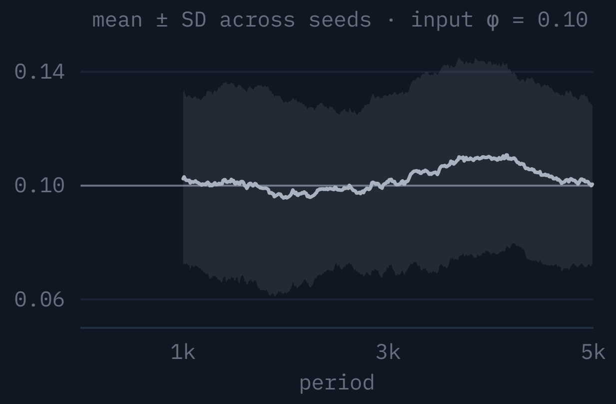
null traders only · seed 42



returns ≈ Gaussian · lag-1 autocorr **0.103** vs input φ = 0.10

No drift without adoption

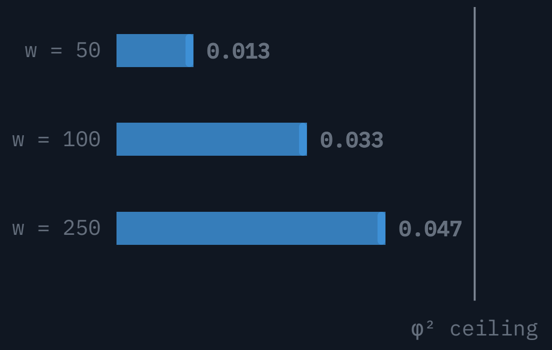
rolling effective φ · 100 seeds



flat in expectation · total range **0.016** over 4,001 periods

The rule finds real signal, offline

no trader acts on it · φ = 0.25



rolling OOS R^2 by window vs the $\varphi^2 = 0.0625$ oracle ceiling

Adoption makes the forecast look better: measured accuracy nearly triples

Out-of-sample R^2 against the realised return

Monte Carlo · 100 seeds · mean ±1 SD by adoption share

REALISED OOS R^2 · ZERO → FULL ADOPTION

0.07 → 0.19

paired shocks, seed 71 · trailing 1,000-period window

Every adopter pushes the same way

Coordinated orders move the realised return **in the forecast's direction**, so the forecast partly scores against its own price impact.

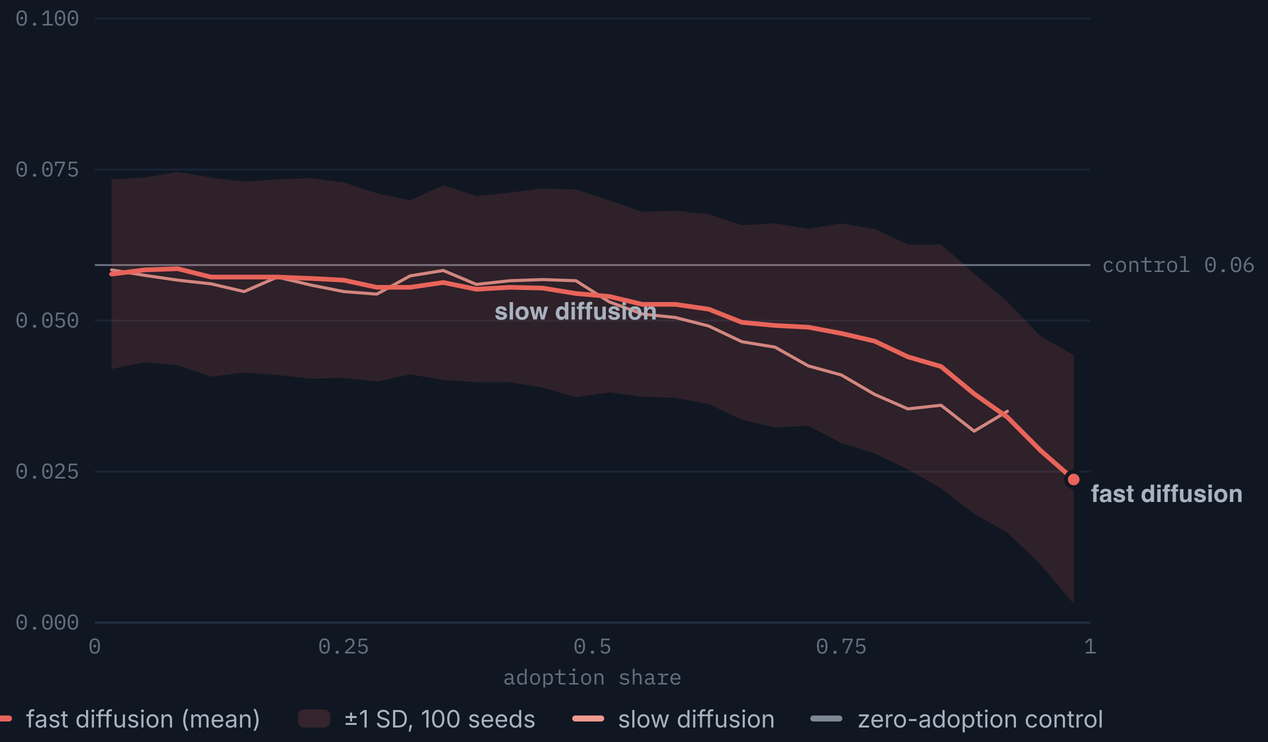
Monotone in adoption, not in time

The zero-adoption control on the same shocks stays flat: the climb is **an adoption effect**, not drift.

On the same paths, the independent signal is quietly dying

Out-of-sample R^2 against the demand-adjusted return

same forecasts, same paths, same seeds · impact stripped from the target



DEMAND-ADJUSTED OOS R^2 · ZERO → FULL ADOPTION

0.08 → 0.04

the same run that printed 0.07 → 0.19 next door

Half the signal's content, gone

Once self-fulfilment is removed from the target, the forecast's grip on the surviving structure **weakens as adoption rises**.

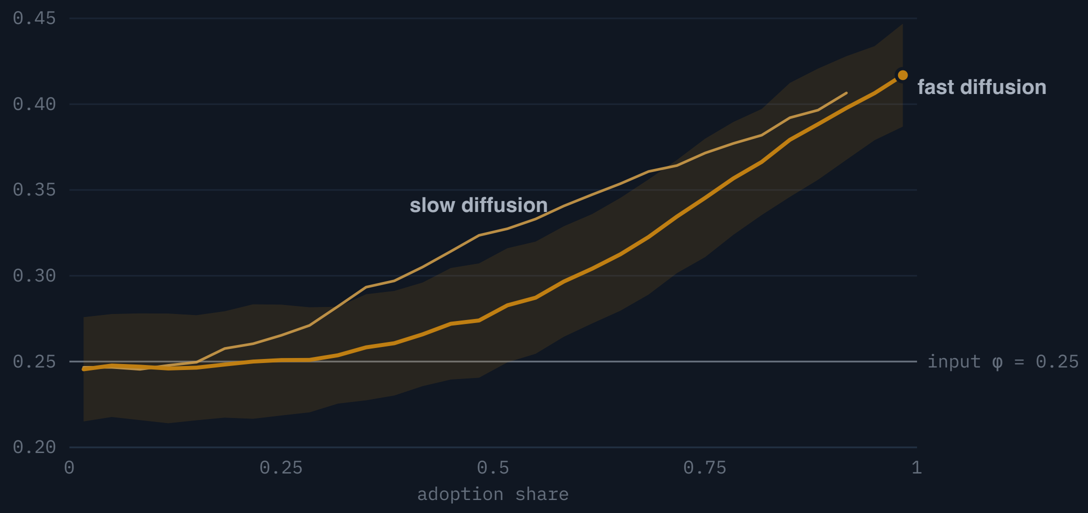
The realised score hides this

Improvement and decay arrive **on the same path, from the same order flow**. The deployed metric shows only the first.

Adopters print extra persistence into the market, and the rolling fit swallows it

Effective AR coefficient of realised returns vs adoption share

rolling lag-1 autocorrelation · 100 seeds · input φ = 0.25 marked



fast diffusion (mean ±1 SD) slow diffusion true ϕ, unchanged by construction

1 · Forecast carries yesterday's sign

An AR(1) forecast is affine in the last return, so adopters trade **the sign of r_t**.

2 · Impact reinforces it

Their block μD_t pushes the next return the same way: **manufactured autocorrelation**, 0.28 → 0.45.

3 · The fit calibrates to the echo

The rolling regression absorbs the amplified coefficient and **over-predicts** the counterfactual, whose true ϕ never moved. That over-prediction *is* the erosion.

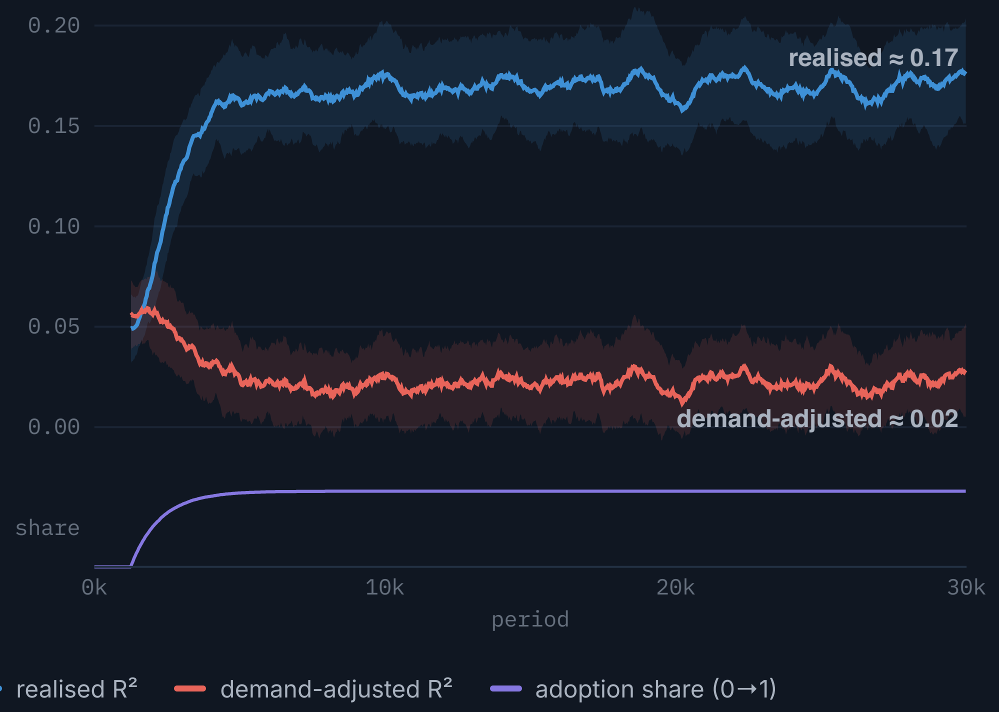
Absorption does not damp it

Market-maker absorption cannot offset it: **μD adds to the next return**, it does not revert it.

Not a transient, and not an artefact of how adoption spreads

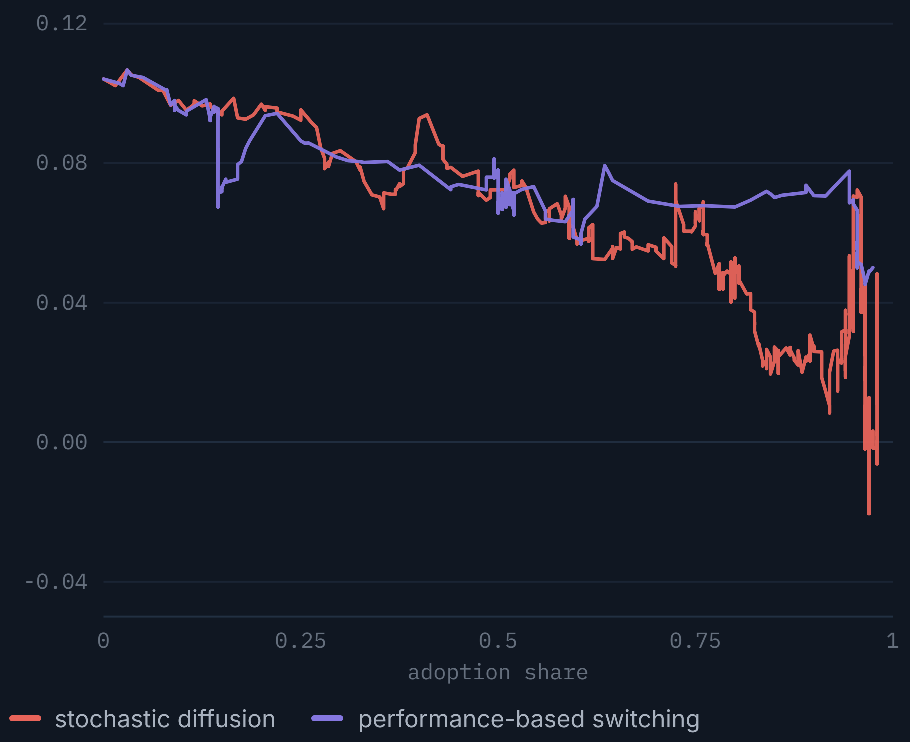
The separation is a steady state

fast diffusion to T = 30,000 · 50 seeds · mean ±1 SD



Endogenous adoption erodes along the same path

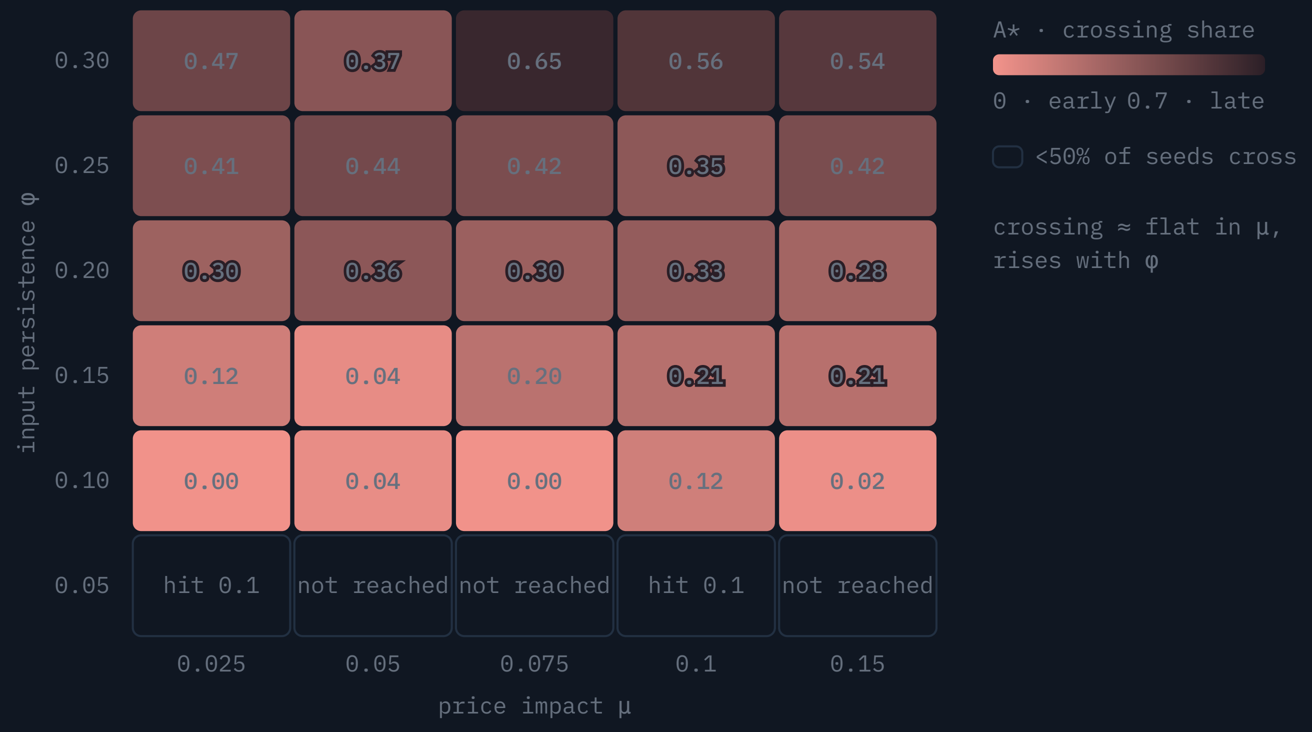
CE switching vs diffusion · paired shocks, seed 71



3,600 runs map *where* the signal loses half its content

Critical adoption share A*: demand-adjusted R^2 falls to half its baseline

AR(1), w = 250 · 10 seeds per cell · brighter = erodes earlier



THRESHOLD REACHED

67 / 90

AR(1) cells · mean detected share 0.29

Stronger signals survive longer

The crossing arrives later as ϕ grows: from near-zero shares at ϕ = 0.10 to **0.37–0.65 at ϕ = 0.30**.

Impact deepens the hole

At the strongest price impact the tail demand-adjusted R^2 is pushed **below zero (to –0.28)**: past halving, to worse than useless.

Survives the forecaster's complexity

Same pattern at AR order p = 1, 2, 5, 10 (**67/59/42/28 cells**); nowhere does it reverse. The realised score never systematically erodes anywhere.

Once a forecast has users, realised accuracy stops measuring skill alone

FOR ANYONE DEPLOYING FORECASTS

— **Realised R^2 over-reports** under adoption: it conflates found signal with the users' own price impact, and improves **exactly while the content erodes**.

— **Report both readings.** The gap between them is the deployment-driven share of measured accuracy, the quantity the headline number silently absorbs.

— The audit needs the period's **order flow D and impact μ**: precisely what an external evaluator usually lacks. That is a data-access argument, not a technicality.

FOR THE THEORY

— Performative prediction's split of accuracy into a **stable part and a deployment-driven part** becomes measurable inside a market with explicit price impact.

— The **demand-adjusted target** identifies the stable component; the threshold maps chart **how much adoption moves each diagnostic by a fixed fraction**.

— Realised evaluation cannot distinguish **a forecast that predicts the market from one the market has come to obey**.

SCOPE, BY DESIGN

Single asset · linear price impact · AR(p) forecasters only · homogeneous null pool. The smallest honest laboratory, not a calibrated market.

WHERE THE MECHANISM NEEDS SIGNAL

At $\phi \leq 0.10$ the baseline is near zero: little signal to lose. Amplification needs both ϕ and μ high enough to bite.

DEFERRED, HONESTLY

The economic endpoint (transaction costs, A^*_{profit}) was attempted; a position-size asymmetry between null and adopter rules must be neutralised first. Future work.

A market that obeys its forecast will grade that forecast generously.

Audit it against the path it did not cause.

THANK YOU · QUESTIONS WELCOME · BACKUP SLIDES FOLLOW

Backup

EQUATIONS · TIMING · THRESHOLD FAMILY · REPRODUCIBILITY · FUTURE WORK

The model in five blocks

MARKET & RETURN LAW

$D_t = (1/N) \sum q_{i,t} \cdot p_{t+1} = p_t + \mu D_t + \sigma\varepsilon_{t+1}$

$r_{t+1} = \varphi r_t + \mu D_t + \sigma\varepsilon_{t+1}$

Market maker absorbs demand levels (Beja-Goldman convention): persistent positions exert persistent pressure.

ORDERS

$z_{i,t} = \hat{r}_{t+1} / a\sigma^2 \cdot q^{(A)} = clip(z, -\bar{q}, +\bar{q})$

Brock-Hommes mean-variance demand; cap binds at baseline, so adopters trade fixed size on the forecast sign. Null: $q^{(0)} \sim N(0, \sigma_q^2)$.

FORECASTING RULE

$\hat{r}_{t+1} = \hat{c} + \sum_i \hat{\varphi}_i r_{t+1-i}$

OLS on the trailing w returns, refit each period, strictly out of sample. p = 1 core; $p \in \{2, 5, 10\}$ robustness.

ADOPTION

$P(0 \to 1) = \pi \cdot P(0 \to 1) = \Lambda(\alpha + \beta S_t)$

Diffusion, or logistic switching in the certainty-equivalent gap $S = CE^{(A)} - CE^{(0)}$ with CE = mean − (a/2)var over $W^A = 500$.

EVALUATION DECOMPOSITION

$r_{t+1} = \mu D_t + x_{t+1} \cdot x_{t+1} = \varphi r_t + \sigma\varepsilon_{t+1} \cdot note\ \sigma\varepsilon_{t+1} = r_{t+1} - \varphi r_t - \mu D_t$

x removes contemporaneous impact only: it is a counterfactual target, not the regression residual. Both R^2 endpoints use the same trailing W = 1,000 window and window-mean benchmark.

The order of events makes the decomposition well defined

1. Agents observe returns **up to and including t** and form the shared forecast $\hat{r}_{t+1}$.
2. Adopters and non-adopters **submit orders** $q_{i,t}$.
3. The market maker absorbs aggregate demand and **moves the quote immediately**: μD_t is in the books.
4. The news shock $\sigma\varepsilon_{t+1}$ and the autoregressive term ϕr_t **realise**.

One critical adoption share per evaluation target

SYMBOL	DEFINITION	WHAT IT DID
$A^*_{R^2,da}$	Rolling demand-adjusted R^2 crosses **zero** (primary, absolute)	Fires only under strong impact or near-zero baseline signal; positive elsewhere
$A^*_{R^2,da,rel}$	Demand-adjusted R^2 falls to **half its own low-adoption baseline**	**The headline threshold:** 67/90 AR(1) cells, mean share 0.29
$A^*_{\varphi,rel}$	Effective AR coefficient reaches **1.5× its baseline**	82/90 cells, mean 0.38; at moderate-to-high ϕ, larger μ pulls it earlier
$A^*_{R^2,realised,rel}$	Realised R^2 falls to half its baseline	The designed non-result: 28/90, all noise crossings at near-zero baselines (26 of 28 at $\phi \leq 0.20$, median baseline 0.014)

Relative factors (0.5, 1.5×) were fixed **before the sweep ran**; each is defined only where the baseline is positive, and a cell reports the mean crossing share only when at least half its 10 seeds cross. Rolling metrics trail adoption by up to the 1,000-period window, so each A* is a detection point: an upper bound on the erosion onset.

Seven phases, each committed before the next began

1–3	**Pre-adoption controls:** baseline market · 100-seed benchmark validation · offline AR rule. Market without forecast trading; forecast without market feedback.
4–5	**Feedback on:** stochastic diffusion, then performance-based switching on paired shocks. The dual channel first appears here.
6	**Quantify:** the (μ, ϕ, w, p) sweep and the critical-share maps.
7	**Evaluate:** dual-channel synthesis; transaction-cost extension attempted, deferred.

Pure numpy, seeded end to end

A 588-line pure-numpy package; every random draw flows through an explicit Generator. **Same seed, same figure.**

43-test suite + CI

Per-phase invariants from seed determinism to cap clipping, run on every push.

Every number traces

Figures save from notebooks with parameters in one top cell; npz summaries back every quoted value, **including the ones on these slides**.

Each direction relaxes one deliberate restriction

The economic endpoint

Proportional cost c per unit traded; adopter net value vs the null benchmark; A^*_{profit}. Attempted: confounded by the position-size asymmetry (null trades $\sim\sigma_q$, adopters capped at $\bar{q}$; the null books own-impact paper profit on a position an order of magnitude larger). Needs a position-matched benchmark and close-out accounting.

Non-linear impact

A square-root impact law would test the mechanism against realistic impact concavity.

Trader ecology

Trend followers and contrarians replacing the random null: other strategies pushing back against the coordinated adopters.

Richer forecasters & assets

Machine-learning rules and multi-asset extensions: does cross-asset demand reproduce, dampen, or redirect the two channels?